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MINIMAL MODEL FOR THERMODYNAMICS OF NONEQUILIBRIUM PHASE TRANSITIONS

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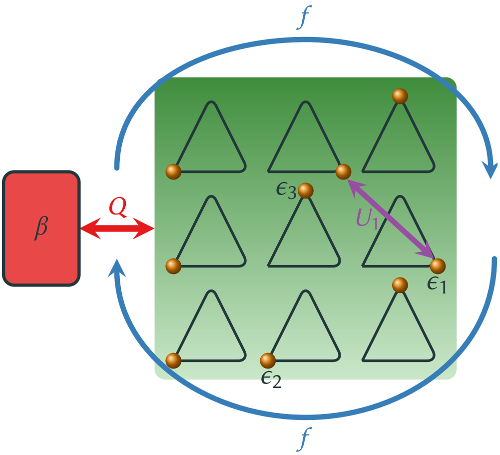
Open Three-State Unit Network

- Network of N all-to-all coupled three-state [PRL **96**, 145701 (2006)] units.

- Open system in contact with heat bath.

- Interaction energy $U(\mathbf{N}) = \frac{u}{2N}\sum_{k=1}^{3} N_k^2 + U_0$.

- Subjected to torque f acting as a global bias.



- Markovian master equation with Arrhenius rates satisfying local detailed balance

$$\dot{P}_{\mathbf{N}} = \sum_{\mathbf{N}'} W_{\mathbf{N},\mathbf{N}'}\, P_{\mathbf{N}'}\,, \qquad \ln \frac{W_{\mathbf{N}\mathbf{N}'}}{W_{\mathbf{N}'\mathbf{N}}} = -\beta\left[\Delta F^{eq}(\mathbf{N},\mathbf{N}') - \sigma(\mathbf{N},\mathbf{N}')f\right], \qquad F^{eq}(\mathbf{N}) = E(\mathbf{N}) - \beta^{-1}\overbrace{S^{int}}^{\ln \frac{N!}{\prod_i N_i!}}(\mathbf{N})$$

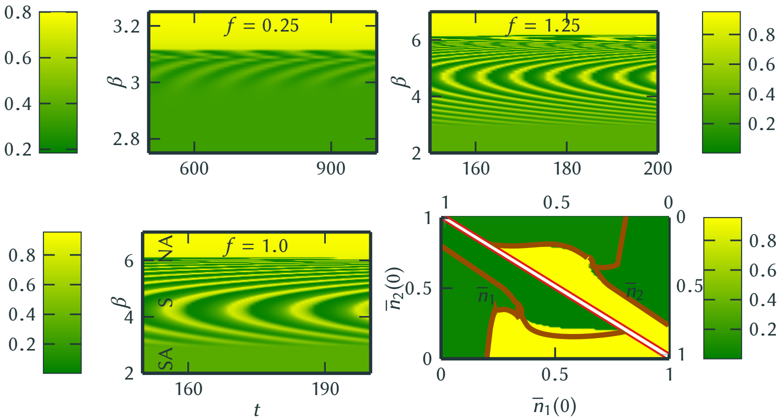
- Mean-field dynamics given by a nonlinear master equation

$$\dot{\overline{n}}_i(t) \equiv \langle n_i(t)\rangle = \sum_j k_{ij}(\overline{n}_i,\overline{n}_j)\,\overline{n}_j(t), \qquad \ln\frac{k_{ij}}{k_{ji}} = -\beta\left[\epsilon_i - \epsilon_j + u(\overline{n}_i - \overline{n}_j) - \sigma(i,j)f\right].$$

- Linear stability analysis around solution $\overline{n}_i^s = 1/3$ indicates a Hopf-Bifurcation at $\beta u = -3$

$$\lambda_\pm = -\Gamma(\beta u + 3)\cosh\left(\frac{\beta f}{2}\right) \pm i\sqrt{3}\,\Gamma \sinh\left(\frac{\beta f}{2}\right).$$

Nonequilibrium $f \neq 0$ and attractive interactions $u \equiv -1$



$\Rightarrow$ Two nonequilibrium phase transitions at $\beta = \beta_{c_{1,2}}$.

- Low- and high temperature regime exhibits same phenomenology in and out-of-equilibrium.
- If sufficiently far away from equilibrium ($f > f_c$), stable oscillations emerge at intermediate temperatures ($\beta_{c_1} < \beta \leq \beta_{c_2}$).

Stochastic Dynamics

- The microscopic/mesoscopic dynamics is ruled by an irreducible and Markovian master equation $\dot{P}_{\mathbf{N}} = \sum_{\mathbf{N}'} W_{\mathbf{N},\mathbf{N}'} P_{\mathbf{N}'}$ and exhibits a unique steady state (PF theorem).

Mean-Field Dynamics

- The macroscopic dynamics is governed by a nonlinear master equation $\dot{\overline{n}}_i = \sum_j k_{ij} \overline{n}_j$ and displays stable oscillations ($\beta_{c_1} \leq \beta < \beta_{c_2}$) and multistability ($\beta \geq \beta_{c_2}$).

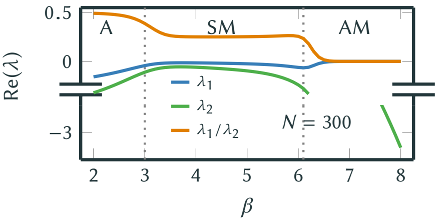
⇒ Non-commutation of the thermodynamic- and the stationary limit.

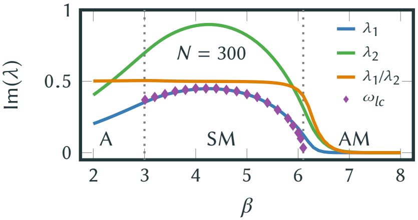
$$\lim_{t\to\infty} \lim_{N\to\infty} \mathbf{P}(t) \neq \lim_{N\to\infty} \underbrace{\lim_{t\to\infty} \mathbf{P}(t)}_{\mathbf{P}^s}, \quad \text{if } \beta \geq \beta_{c_1}.$$

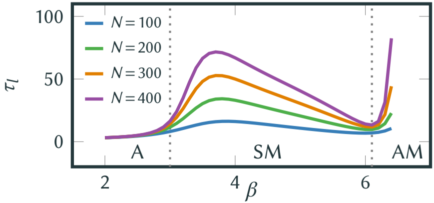
- Apparent paradox since mean field represents the exact solution in the $N \to \infty$ limit.
- Idea: Infinite-time solutions of mean-field exhibited also by finite systems at long but *finite* timescales → Metastability?

Spectral Analysis

- Formal solution $\mathbf{P}(t) = e^{\mathbf{W}t}\,\mathbf{P}(0) = \sum_{i,i^*} \underbrace{\Phi_i^L\,\mathbf{P}(0)}_{\equiv c_i}\, e^{\lambda_i t}\,\Phi_i^R + c_{i^*}\,e^{\lambda_{i^*} t}\,\Phi_{i^*}^R$.
- Time-scales are encoded in $\mathbf{W}$: $\tau_r \sim -\frac{1}{\mathrm{Re}(\lambda_1)}$, $\quad \tau_m \sim -\frac{1}{\mathrm{Re}(\lambda_2)}$, $\quad \tau_l \equiv \tau_r - \tau_m \sim \frac{\mathrm{Re}(\lambda_1)-\mathrm{Re}(\lambda_2)}{\mathrm{Re}(\lambda_1)\,\mathrm{Re}(\lambda_2)}$.

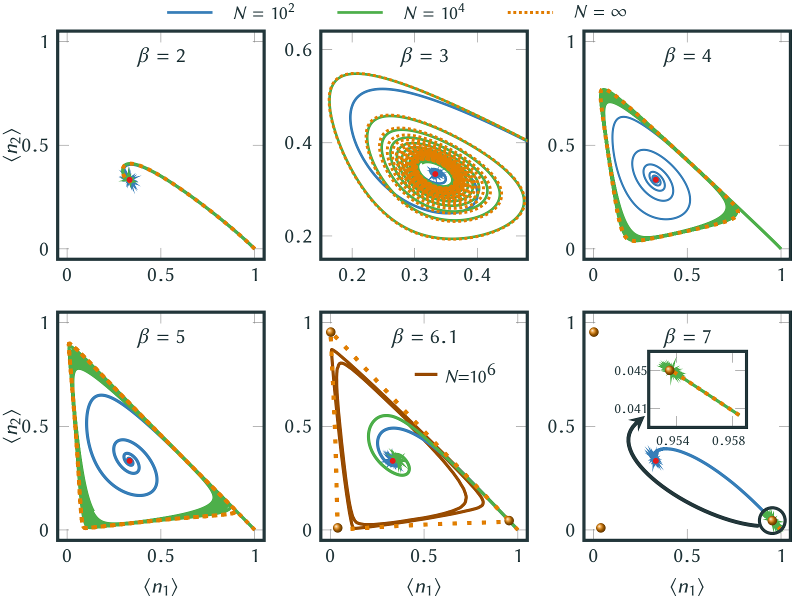






⇒ Relaxation dynamics at metastable times ($\tau_m \ll t \ll \tau_r$) are characterized only by the modes constituting the metastable state, *i.e.*

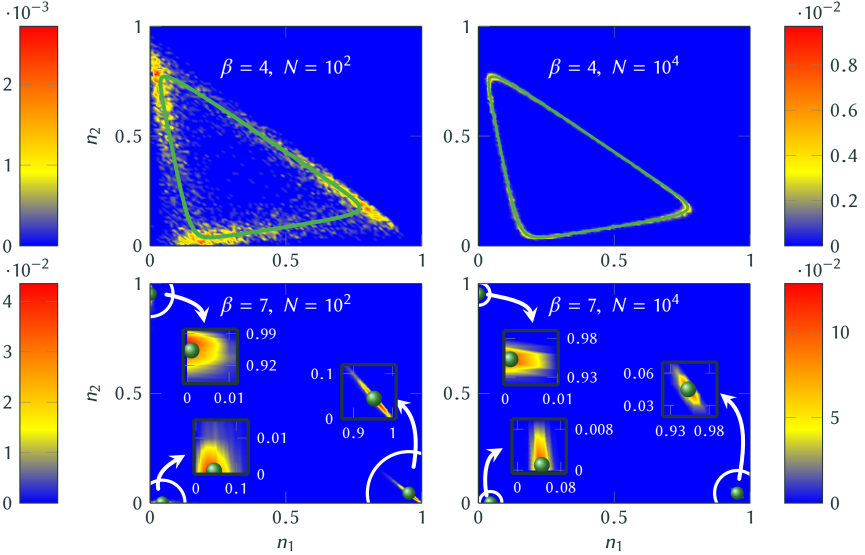
$$\mathbf{P}(t) \approx c_0\Phi_0^R + e^{\lambda_1 t}c_1\Phi_1^R + e^{\lambda_1^* t}c_1^*\Phi_1^{R*}$$



$\Rightarrow$ Large system dynamics indeed agrees with the mean-field solution at metastable times.

- Recall that $\mathbf{P}(t) \overset{t \gg \tau_m}{\approx} c_0 \Phi_0^R + e^{\lambda_1 t} c_1 \Phi_1^R + e^{\lambda_1^* t} c_1^* \Phi_1^{R^*}$ with $c_{1,1^*} = \Phi^L_{1,1^*} \cdot P(0)$.
- Metastability can be removed from the dynamics for $P(0) = b_0\, \Phi_0^{R^*} + \sum_{i \neq 1, 1^*} b_i\, \Phi_i^{R^*}$.

Is metastability a generic property of our system or merely an artifact of special initial conditions?



Role of initial condition for Metastability II

$\beta = 4$

$N = 10^2$

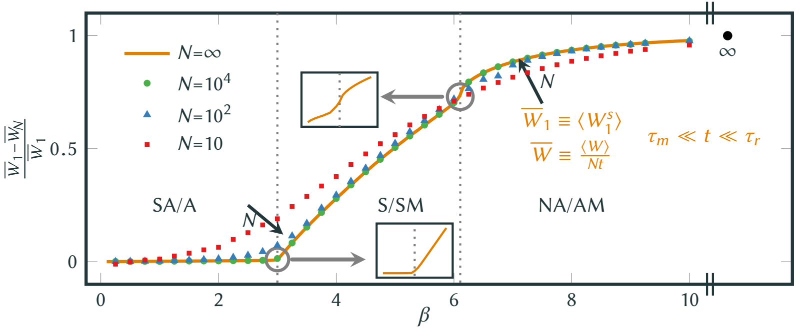
$N = 10^4$

$\beta = 7$

$N = 10^2$

$N = 10^4$

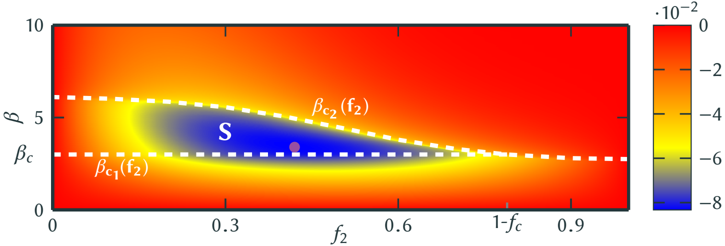
- With this setup, work needs to be done on average to maintain the system, $\langle \dot{W} \rangle > 0$, which is on average dissipated as heat into the bath, $\langle \dot{Q} \rangle < 0$.

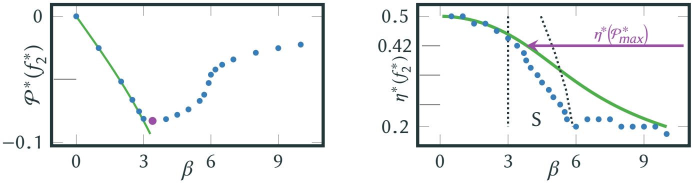


- (Signatures of) Nonequilibrium phase transitions at the critical points (for large but finite N). ["First-order phase transition" at β_{c_1} and "second-order phase transition" at β_{c_2}].
- Reduction of the dissipated work for finite interactions $u < 0$.
- Dissipated work is further reduced for smaller networks in the two high-temperature regimes.

Efficiency at maximum power

- Work-to-work conversion $f \equiv f_1 - f_2$, with input work $\mathcal{W}_1(f_1)$ and output work $\mathcal{W}_2(f_2)$.
- Maximization of the output power $\mathcal{P} \equiv \dot{\mathcal{W}}_2$ with respect to output force $\partial_{f_2}\mathcal{P}\big|_{f_1=1} = 0 \Rightarrow f_2^*$.
- Efficiency of work-to-work conversion at maximum power $\eta^* \equiv 1 - \frac{f}{f_1}\Big|_{f_2=f_2^*}$





- Power output is maximized when the network is synchronized, *i.e.* sufficiently far driven out-of-equilibrium $f \geq f_c$.
- The efficiency at maximum power, $\eta^*(f_2^*)$, is surprisingly close to the universal value, $\eta^* = 1/2$, for a tightly-coupled (single net-current) system responding linearly.

Thank you for your attention!

Collective power: Minimal model for thermodynamics of nonequilibrium phase transitions

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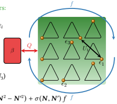
Abstract

We establish a direct connection between the linear stochastic dynamics, the nonlinear mean-field dynamics, and the thermodynamic description of a minimal model of driven and interacting discrete oscillators [1,2]. This system exhibits at the mean-field level two bifurcations separating three dynamical phases: a single stable fixed point, a stable limit cycle indicative of synchronization, and multiple stable fixed points. These complex emergent behaviors are understood at the level of the underlying linear Markovian dynamics in terms of metastability. Thermodynamically, the dissipated work of the stochastic dynamics exhibits signatures of nonequilibrium phase transitions over long metastable times which disappear in the infinite-time limit. Remarkably, it is reduced by the attractive interactions between the oscillators. When operating as a work-to-work converter, we find that the maximum power output is achieved far-from-equilibrium in the synchronization regime and that the efficiency at maximum power is surprisingly close to the universal linear regime prediction.

Model

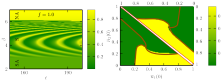
Minimal model of driven oscillators:

- Three-state units with energies ϵ_i
- Heat bath at temperature β
- Nonconservative force f
- **Markovian master equation**
  $P_{\boldsymbol{N}} = \sum_{\boldsymbol{N}'} W_{\boldsymbol{N},\boldsymbol{N}'} P_{\boldsymbol{N}'}$, $\boldsymbol{N} \equiv (N_1, N_2, N_3)$
- **Local detailed balance**
  $-\frac{1}{\beta}\ln\frac{W_{\boldsymbol{N},\boldsymbol{N}'}}{W_{\boldsymbol{N}',\boldsymbol{N}}} = (\boldsymbol{N}-\boldsymbol{N}')\cdot\boldsymbol{\epsilon} + \frac{u}{2N}(\boldsymbol{N}^2 - \boldsymbol{N}'^2) + \sigma(\boldsymbol{N},\boldsymbol{N}')\, f$



Mean-Field Dynamics

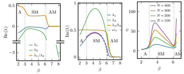
- **Dynamics:** $\dot{\bar{n}}_i = \sum_j k_{ij}(\bar{n}_i, \bar{n}_j)\,\bar{n}_j$, $-\frac{1}{\beta}\ln\frac{k_{ij}}{k_{ji}} = \epsilon_i - \epsilon_j + u(\bar{n}_i - \bar{n}_j) + \sigma(i,j) f$.
- SA phase: For symmetric units, $\epsilon_i = \epsilon$, the dynamics goes to $\bar{n}_i^s = 1/3$.
- S phase: $\lambda_{1,2} = -(\beta u + 3)\cosh\left(\frac{\beta f}{2}\right) \pm i\sqrt{3}\sinh\left(\frac{\beta f}{2}\right) \Rightarrow$ Hopf bifurcation at β_{c_1}.
- NA phase: Infinite-period bifurcation at β_{c_2}.



- **Two** bifucations separating **three dynamical phases**:
  - Unique symmetric fixed point for $\beta < \beta_{c_1}$
  - Limit cycle for $\beta_{c_1} \le \beta < \beta_{c_2}$
  - Three asymmetric fixed points for $\beta_{c_2} \le \beta$
- No oscillations in the low- and high-temperature limit due to **consistency**.
- $\Rightarrow$ **Thermodynamically consistent minimal model for synchrony** .

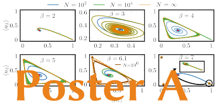
Metastability

- Unique stationary state $\langle n_i^s \rangle = 1/3$, *i.e.* $\lim_{t\to\infty}\lim_{N\to\infty} P_{\boldsymbol{N}} \neq \lim_{N\to\infty}\lim_{t\to\infty} P_{\boldsymbol{N}}$, if $\beta \ge \beta_{c_1}$.
- **Timescales**: $\tau_r \sim -\frac{1}{\mathrm{Re}(\lambda_1)}$, $\tau_m \sim -\frac{1}{\mathrm{Re}(\lambda_2)}$, $\tau_l \equiv \tau_r - \tau_m \sim \frac{1}{|\mathrm{Re}(\lambda_1)|}\left(1 - \frac{\mathrm{Re}(\lambda_1)}{\mathrm{Re}(\lambda_2)}\right)$.

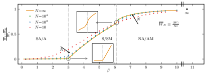


- **Time-scale seperation** between the fast ($i \ge 2$) and the slowest finite-time ($i = 1$) and the stationary mode ($i = 0$) $\Rightarrow$ **Metastability**.
- **Metastable states** correspond to the mean-field solutions at $\tau_m \ll t \ll \tau_r$.
- Smooth convergence to mean field since $\tau_l, \tau_r \to \infty$ as $N \to \infty$.

Stochastic Dynamics

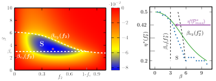


Disssipated Work



- "First"- and "second-order-like" nonequilibrium **phase transitions** at $\beta_{c_{1,2}}$.
- Signatures of **phase transitions** for finite systems at **metastable times**.
- **Reduction of dissipated work** for any finite attractive interaction.

Efficiency-Power Tradeoff



- **Maximum power** in the far-out-of-equilibrium **synchronization** phase.
- **Efficiency at maximum power** is quite close to **linear response** value.

References

[1] T. Herpich, J. Thingna, and M. Esposito, arXiv:1802:00461 (2018).
[2] K. Wood, et al., Phys. Rev. Lett. **96**, 145701 (2006).

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